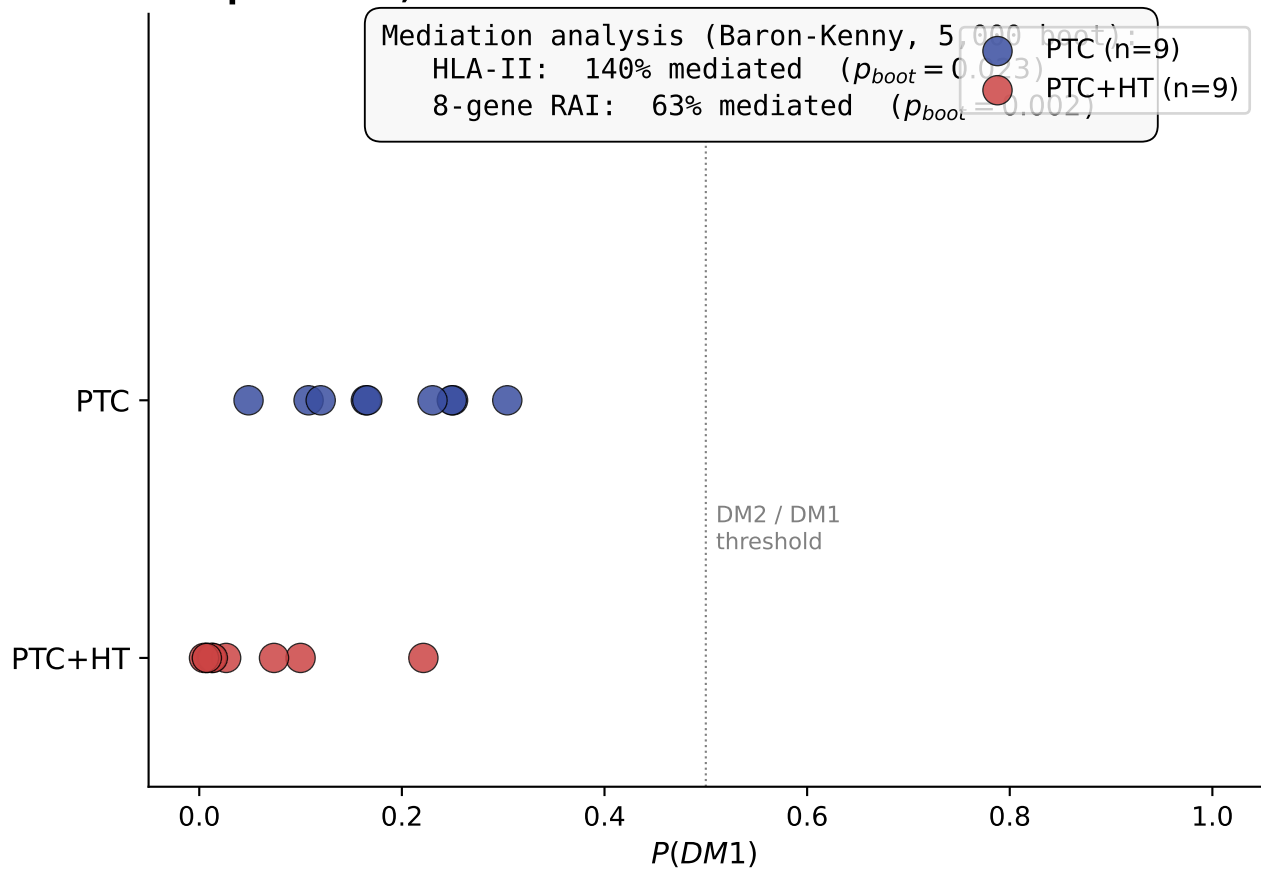
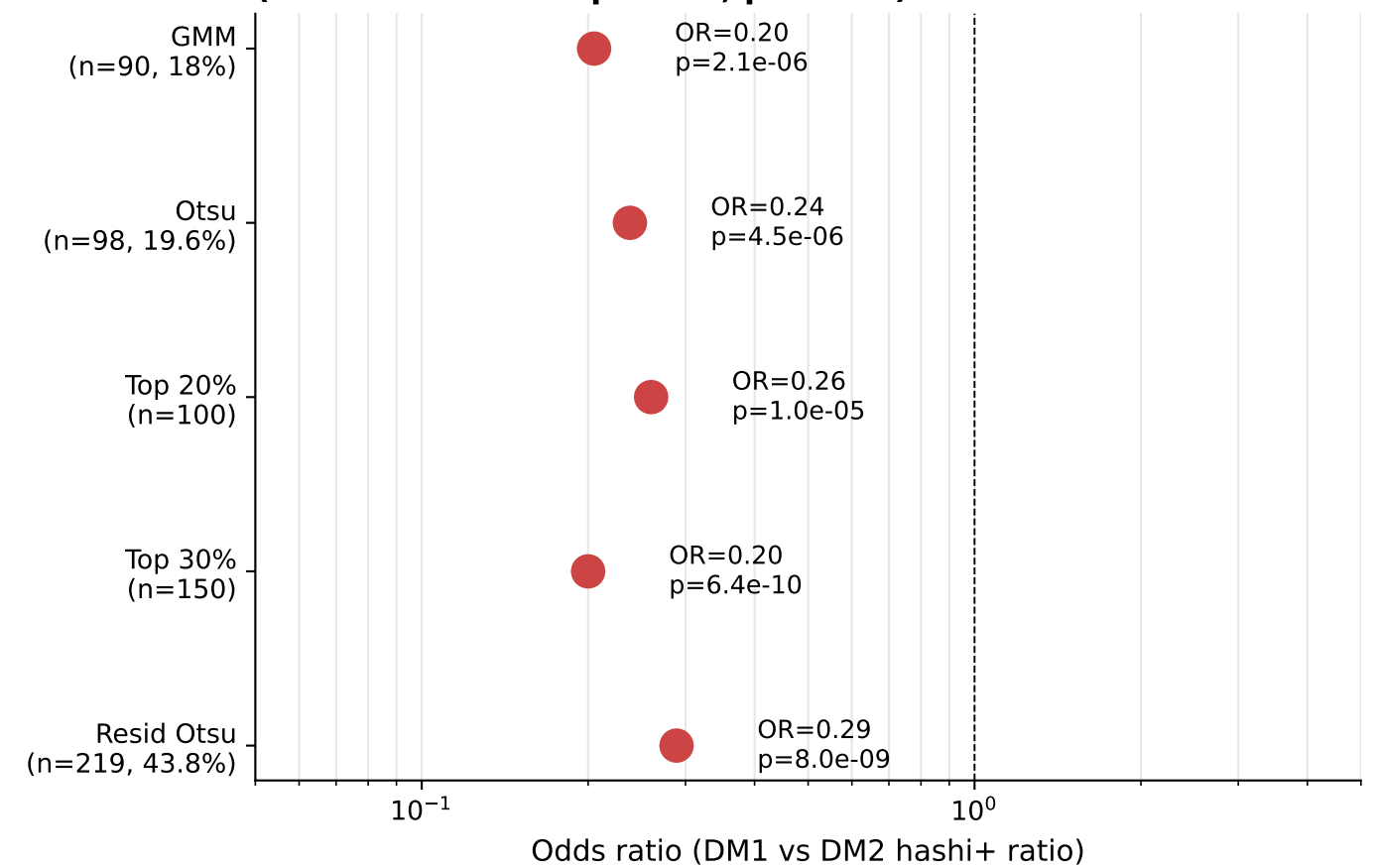


# Figure 5 — Autoimmune-PTCmechanism layer (Pillar 5)

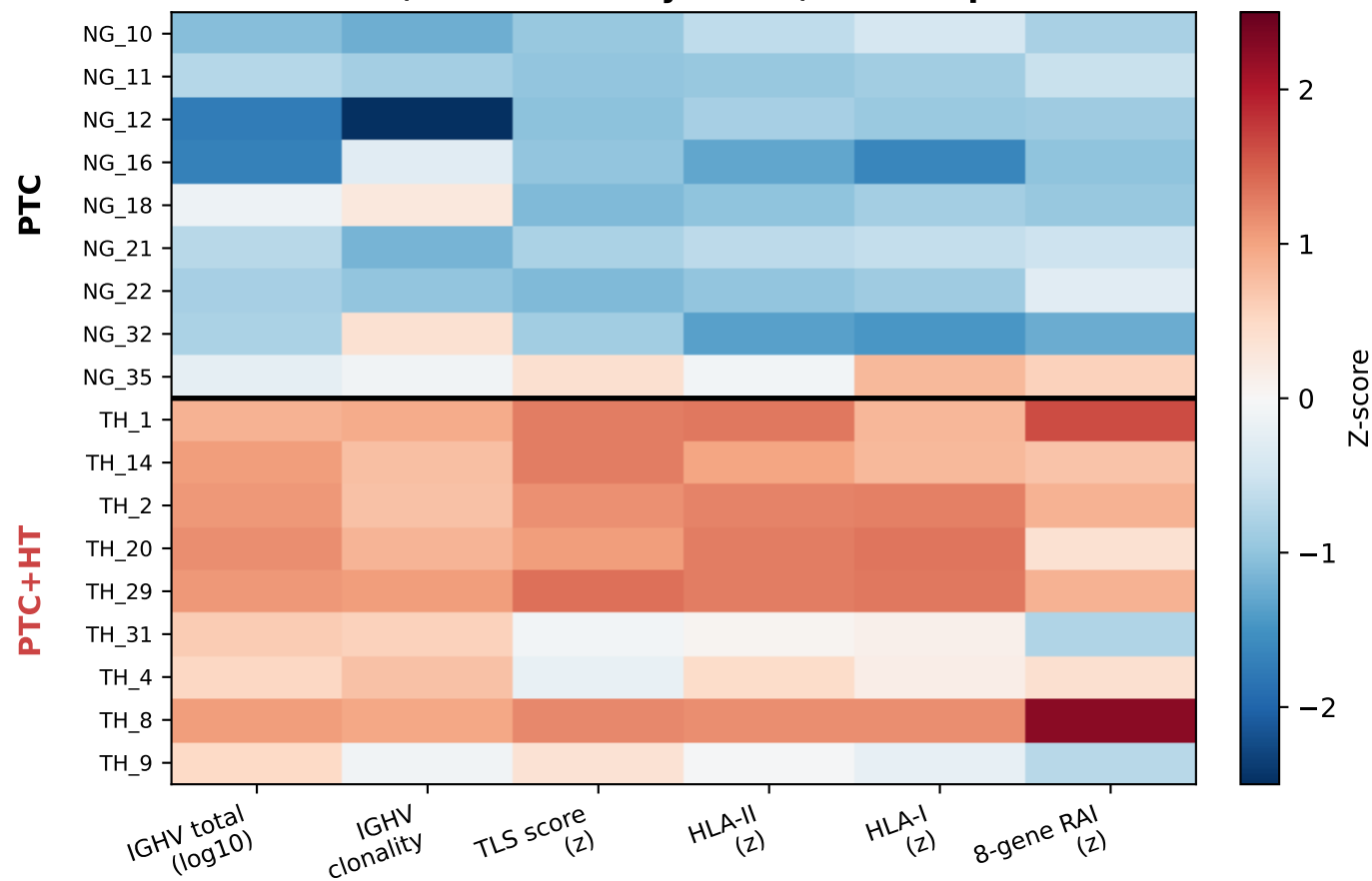
**A — P(DM1) spectrum (GSE286332 n=18)**  
MW p=0.0036; HLA-II is dominant mediator



**B — TCGA-THCA Hashimoto-like signature × DM cluster**  
(★ DM2 enriched up to 5×, p=6e-10)



**C — BCR repertoire + TLS + HLA per sample (n=18)**  
TLS d=+1.96, IGHV clonality d>0.5, AICDA up



**D — DM1 sub-cluster mutation × Hashimoto stack**  
sub-A: 69% RAS+ FVPTC core; sub-B: 96% mut-neg NBNR

